

Lab 05

Collector Feedback Bias and Voltage Divider Bias BJT Biasing Configurations

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5.1 Objective

To investigate the effect of current gain β variation on quiescent operating conditions of a Bipolar Junction Transistor in collector feedback bias and voltage divider bias configurations.

Note: Pre-lab (printed or hand-written) should be done on a separate paper (use A4 sheet) and will be submitted at the start of Lab within the first 5 minutes.
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5.2 Introduction

In previous lab, the stability of fixed-bias and emitter-bias BJT biasing configurations have been investigated. In this lab, you will investigate the dependency of BJT operating point on DC current gain β for two more circuits of BJT biasing; collector feedback bias (with and without emitter resistor) and voltage divider bias configuration.

5.3 Pre-Lab Task

Task 1:

i. List down two differences between 2N3904 and 2N4401 in terms of power rating and beta value.

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| <p>ii. The 2N3904 and 2N4401 transistors mainly differ in their power/current handling capability and their DC current gain (β).</p> <p>iii. First, in terms of collector current (and power rating), the 2N4401 can handle much more current. It has a maximum collector current of about 600 mA, while the 2N3904 is limited to around 200 mA. Because of this, the 2N4401 is more suitable for circuits where higher current or slightly higher power is involved, whereas the 2N3904 is typically used in low-power, small-signal applications.</p> <p>iv. Second, their beta (β) range is slightly different. The 2N4401 generally has a higher and wider β range (about 100–400) compared to the 2N3904 (about 100–300). This means the 2N4401 can provide a bit more current amplification under the same biasing conditions, although in practical circuits both are often designed to work properly despite β variations.</p> |
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v. State advantages and disadvantages of the following transistor biasing configurations and which configuration is most used and why?

1. Fixed Bias Configuration
2. Common Emitter Bias configuration
3. Collector Feedback Bias configuration
4. Voltage Divider Bias configuration

1. Fixed Bias Configuration

Advantages:

- Very simple circuit with minimum components (only one base resistor).
- Easy to design and analyze.

Disadvantages:

- Highly dependent on the transistor's β ; small changes in β shift the Q-point significantly.
- Very poor thermal stability, increase in temperature increases collector current, which can drive the transistor into saturation.

2. Common Emitter Bias (Emitter-Stabilized Bias)

Advantages:

- Better thermal stability than fixed bias due to the emitter resistor providing negative feedback.
- Q-point is more stable with temperature changes.
- Simple improvement over fixed bias (only one extra resistor required).

Disadvantages:

- Still somewhat dependent on β .
- The emitter resistor reduces AC voltage gain if it is not bypassed.
- Using a bypass capacitor increases cost and circuit complexity.

3. Collector Feedback Bias Configuration

Advantages:

- Provides automatic stabilization: if I_C increases, V_C decreases, which reduces I_B and brings I_C back down.
- Simple and requires only a few components.

Disadvantages:

- AC feedback from collector to base can reduce voltage gain.
- Stability is better than fixed bias but not as good as voltage divider bias.
- Still affected by β variations to some extent.

4. Voltage Divider Bias Configuration

Advantages:

- Most stable biasing method.
- Base voltage is set by the voltage divider and is almost independent of β (for a stiff divider).
- Q-point remains stable with temperature and transistor replacement.
- Gives predictable and reliable performance.

Disadvantages:

- Requires more components (two base resistors + emitter resistor).
- Continuous current flows through the divider, causing slight power loss.
- Design calculations are a bit more involved.

Most Commonly Used Configuration and Why

The **voltage divider bias configuration** is the most widely used in practical amplifier circuits because:

- The operating point is almost independent of β .
- It provides excellent thermal and Q-point stability.
- It works reliably even if the transistor is replaced.
- It is ideal for real-world and mass-production applications where device parameters vary

